

Team Worksheet — Lab 20

Multi-Vari Studies and Chi-Square Analysis (Black Belt Only)

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-20-multivari-chisquare.xlsx`.

3. What we produced

Deliverable: A multi-vari chart decomposing your capstone variation plus a chi-square association test.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Understand the three variation families. POSITIONAL: within a single unit or across positions in a machine. CYCLICAL: unit to unit, batch to batch. TEMPORAL: shift to shift, day to day.

Step 2. Understand the payoff: if 70 percent of variation is positional, then investigating shift patterns is wasted effort no matter how thorough the investigation.

Step 3. Design a multi-vari sampling plan for your capstone: sample multiple positions within units, consecutive units, and repeat across time periods.

Step 4. Ensure the sampling captures all three families simultaneously — this is a nested sampling design, not three separate studies.

Step 5. Collect the data and plot the multi-vari chart, with positional variation innermost and temporal outermost.

Step 6. Read the chart: the family showing the largest vertical spread dominates. That is where your root cause investigation should concentrate.

Step 7. Quantify the contribution of each family as a percentage of total variation using variance components.

Step 8. Re-prioritise your candidate cause list from Lab 16 against the dominant variation family, and drop causes that belong to a family contributing little.

Step 9. Now chi-square. Build a contingency table of two categorical variables from your capstone — defect type against shift, supplier or line.

Step 10. Calculate expected frequencies under independence: $\text{expected} = (\text{row total} \times \text{column total}) / \text{grand total}$.

Step 11. Check the validity condition: all expected frequencies should be 5 or more, or the chi-square approximation breaks down and categories must be combined.

Step 12. Compute the chi-square statistic and its p-value, and conclude whether defect type is associated with your categorical factor.

Step 13. Translate a significant association into a testable process hypothesis — association is not causation, but it does tell you where to look.

7. Self-check

Your multi-vari chart captures all three families with quantified variance contributions, your cause list is re-prioritised against the dominant family, and your chi-square satisfies the expected-frequency condition.

Complete? ■ Yes ■ Not yet

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